

# Tracking and Removing a Migrating Window Blemish

CMU Skyline Monitor — pandr / akdeniz

2026-07-28 (v8)

## Abstract

A  $\sim 0.25$  mm speck on the window of a sky monitor is imaged by the camera's  $f/4$  pupil from  $\sim 3$  cm as a 76 px, 2 % to 7 % shadow, which contrast enhancement amplifies  $\sim 2.9\times$  in the served image. The shadow geometry acts as a  $\sim 92$  px/mm strain gauge: sub-millimetre thermal flexure of sash and mount walked the feature's image 120 px in a week, including 7 px to 18 px within single days. We remove it with a layered Bayesian scheme: a hierarchical pixel model (Student- $t$  likelihood, position pinned across exposure brackets, days tied by a drift-informed random walk) locates the feature; a nightly recursive filter carries position and velocity forward, each estimate stamped with its data's epoch; and at every capture a template-lock measurement update corrects the prediction, with depth taken from a registered empirical transmittance map rather than any parametric shape. On clean daylight frames with a detectable dip, the median displayed artifact falls from 4.5 % to 0.5 % (day-blocked 95% CI on removal: 86 % to 91 %).

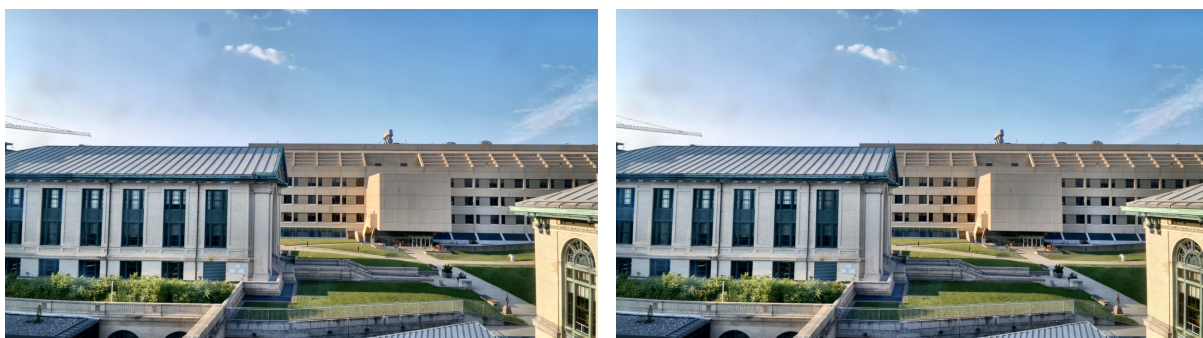


Figure 1: The same three raw brackets fused without (left) and with (right) the correction — 2026-07-25 19:17 EDT. The blemish is the dark smudge at the top of the sky, left of centre; displayed dip 5.46 %  $\rightarrow$  -0.5 % (nulled to within measurement noise). The corrected fusion is the production chain's output.

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## 1 The instrument and the defect

A 4K camera (3840×2160) images the CMU skyline through a fixed window. Every five minutes it captures a three-exposure bracket, fused by Mertens exposure fusion and finished with a local-contrast pass (CLAHE) plus global saturation and contrast curves. Brackets are archived losslessly, so all photometry runs on raw frames; the correction is applied to the raw brackets *before* fusion, because the enhancement stage amplifies any residual dip roughly 2.9×, scene-dependently. The camera (Elgato Facecam 4K) is an  $f/4.0$  fixed-focus design — Elgato Prime Lens, 21 mm equivalent on a 1/1.8" sensor (actual focal length  $\approx 4.3$  mm, entrance pupil  $\approx 1.1$  mm), focus range 30 cm to 120 cm — mounted with its lens under 1 cm from the pane. These numbers matter: the blemish is imaged as an out-of-focus pupil shadow, and every inference about its physical nature runs through them (§9). The blemish currently projects to  $\sim (1300, 132)$  — only 130 px below the top edge of the sensor, which constrains the correction geometry (§5).

**What is deployed and what is offline.** This note describes code in three states, and labels each claim accordingly: *deployed* (running on the capture host `pandr` or the compute host `akdeniz` as of 2026-07-28), *offline* (analyses run to produce the results here, with code retained under `spot_paper/code/`), and *proposed*. The authoritative component-by-component table, with byte-copies of the deployed files, is `PRODUCTION_STATUS.md` and `production/` in the project directory. An earlier draft did not make these distinctions and correctly drew a reviewer’s objection.

## 2 Statistical model and inference

### 2.1 Photometric model

Per colour channel, in linear light, the blemish maps the background radiance it occludes as an affine transformation,

$$L_{\text{sp}} = t L_{\text{bg}} + s, \quad \text{so the fractional dip} \quad d = 1 - L_{\text{sp}}/L_{\text{bg}} = (1 - t) - s L_{\text{bg}}^{-1}. \quad (1)$$

The intercept of  $d$  against  $L_{\text{bg}}^{-1}$  identifies the transmittance  $t$ ; the slope is exactly  $-s$ , so a nonzero slope is a direct test for an additive component (reported in §5). One structural fact shapes everything: *an exposure bracket cannot separate  $t$  from  $s$* . Changing exposure scales  $L_{\text{sp}}$  and  $L_{\text{bg}}$  together, so within a bracket the three points are collinear with slope  $t + s/L_{\text{bg}}$ . Brackets calibrate the tone curve, nothing more; separating  $t$  from  $s$  requires natural variation of  $L_{\text{bg}}$  at fixed window irradiance (cloud behind the blemish, the twilight ramp).

### 2.2 Locating the blemish: spatial models (offline analysis)

Equation (1) uses scalar aperture summaries and carries no spatial information. Two spatial estimators exist, and it matters which is which. The *production* nightly estimator (deployed, `spotnight.py`) is deliberately simple: median-stacked ratio maps, position pinned across brackets, a half-depth centroid, and a recursive prior with step and spread gates. The *Bayesian pixel model below is the offline analysis* that validates it: fitted independently, the two agree along

the whole eight-day track to  $\sim 3$  px, which is the justification for running the cheap estimator nightly. In the offline model, a level-normalised image window is a smooth background minus a blob,

$$y_i \sim \text{Student-}t\left(\nu, b_{0,m} + \mathbf{x}_i^\top \boldsymbol{\beta}_m - A_{k(m)} e^{-Q_i/2}, \sigma_m\right), \quad (2)$$

with  $m$  indexing image stacks,  $k(m)$  the exposure bracket of stack  $m$  (one amplitude per bracket class),  $\mathbf{x}_i$  the quadratic basis  $(u, v, u^2, uv, v^2)$  standardised per stack, and

$$Q_i = \frac{1}{1 - \rho^2} \left[ \frac{(u_i - c_x)^2}{\sigma_x^2} - \frac{2\rho(u_i - c_x)(v_i - c_y)}{\sigma_x \sigma_y} + \frac{(v_i - c_y)^2}{\sigma_y^2} \right],$$

with priors  $c_x, c_y \sim \mathcal{N}(0, 30)$  px about the drift-predicted centre and a non-centred random walk of scale  $\tau$  px/night tying days;  $\sigma_{x,y} \sim \mathcal{N}(33, 7)$  bounded to  $[15, 60]$  px;  $\rho \sim \mathcal{N}(0, 0.35)$  on  $[-0.85, 0.85]$ ;  $A_k \sim \mathcal{N}(0.02, 0.012)$  bounded to  $[0.002, 0.10]$  (a positive floor, see below); Student- $t$  errors ( $\nu \sim \text{Gamma}$ ,  $\nu \geq 2$ ) keep cloud edges from dragging the centre. Exact code, bounds, seeds and the multi-start command lines are in `spot_paper/code/`. Three structural choices carry the statistical weight:

- **One position, shared across brackets.** The three brackets of a tick see the same glass at the same instant. Pinning the centre across brackets (after normalising each bracket’s dip by its own peak, so shape is compared rather than depth) reduced per-bracket disagreement from  $\sim 60$  px to a few px.
- **Positions tied across days by a drift-informed random walk.** Coordinates are expressed relative to each day’s predicted centre, so the parameters are small deviations,  $c^{(d)} = c^{(d-1)} + \tau \eta_d$ ,  $\eta_d \sim \mathcal{N}(0, 1)$ , with  $\tau$  a few px: the geometry changes smoothly, it does not teleport.
- **Depth free per exposure; shape shared.** Apparent depth is tone-curve dependent; the blob’s location and width are physical. A depth that scales with exposure at a fixed location is a signature a smooth background cannot mimic, and it is what identifies the blob against the background polynomial.

Identifiability requires care rather than luck: the background basis must be standardised per stack, the blob width bounded well below the window half-width (a blob as wide as its window *is* a background quadratic), and the amplitude given a positive floor (at  $A = 0$  the position has no likelihood gradient and reverts to its prior). With those constraints, MAP estimation (CmdStan L-BFGS, multiple starts, best objective kept) recovers the position in seconds. The output is a *MAP state*, not a posterior — the posterior is never sampled — and the trust decision is made by an acceptance gate (§7) rather than by a credible interval. The fitted half-depth radius, 37.8 px, independently matches the 38.2 px measured by a completely separate empirical estimator.

### 2.3 A recursive filter with honest epochs

For unattended operation the nightly fit is a filter: tonight’s prior is last night’s posterior propagated by the blemish’s own observed velocity (self-calibrating — no parallax constant, no scene-drift proxy), with implausible steps and low-confidence fits rejected in favour of carrying the previous state forward.

One discipline deserves emphasis because everything downstream depends on it: **an estimate carries its data’s epoch, not its run date.** The nightly position is fitted from the two previous days of frames, so its effective epoch is the mean timestamp of those frames ( $\approx 1.5$  days before the run). The estimator stores that epoch in its state, velocity is the centre step divided by the *elapsed time between epochs* (px/day — robust to skipped nights, gaps and replays), and the publisher stamps the epoch into the model bundle so all placement extrapolation is

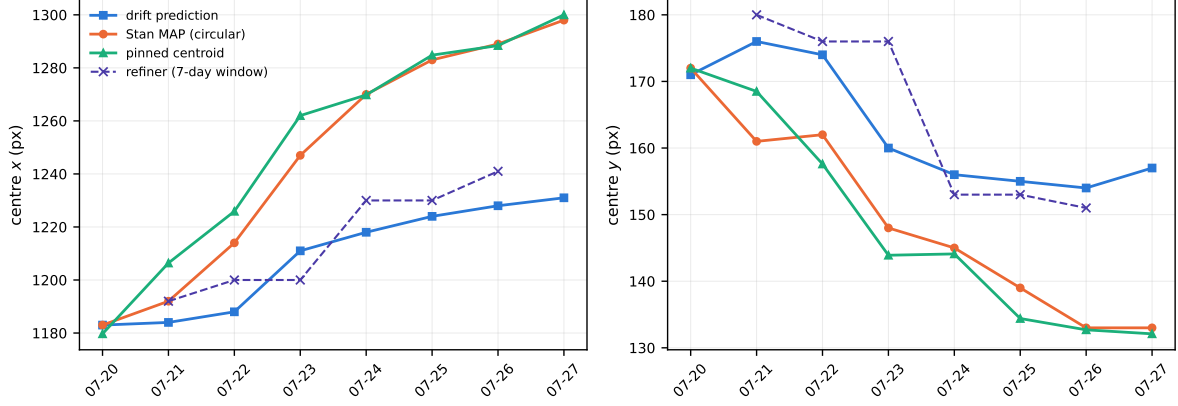


Figure 2: Estimates of the blemish centre across eight days. The scene-drift prediction systematically lags: blemish motion is *not* proportional to scene motion (§5). The Bayesian MAP track and the independent pinned-centroid track agree to  $\sim 3$  px at both ends while sharing no machinery.

referenced to it. Mislabelling the epoch by a day shifts every placement by  $v \Delta t$  and prints a bright/dark dipole at the blemish — which is how the requirement was learned, twice: first as a run-date-vs-data-date confusion, then as a reviewer’s observation that the state carried no epoch at all.

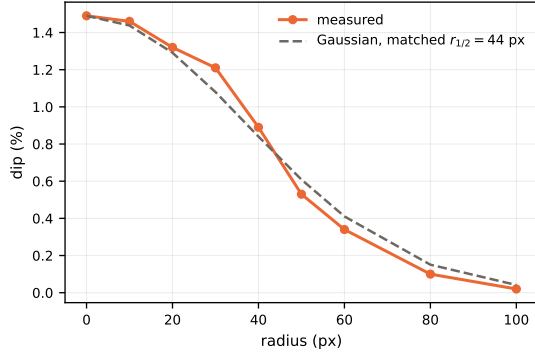
### 3 The empirical transmittance map

The blemish’s radial profile is not Gaussian: its wings fall faster than a Gaussian matched at half depth (Figure 3a). The parametric blob is therefore used only to *locate*; depth and shape come from an empirical map,

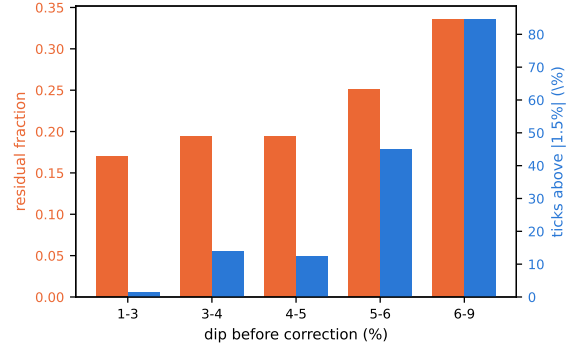
$$T_c(\mathbf{r}) = \text{median}_j \frac{I_{j,c}(\mathbf{r})}{B_{j,c}(\mathbf{r})}, \quad (3)$$

the per-channel median, over selected frames, of each frame divided by its own quadratic background, in linear light. Moving cloud averages out; fixed attenuation survives. Four details matter:

- **Registration.** Each frame’s window is extracted at that frame’s velocity-interpolated centre, so the dip stacks aligned instead of being smeared by its motion across the stacking window.
- **Blemish-local, sigma-clipped background rings.** The background is fitted on an annulus centred on the *blemish* (never the window centre), with MAD-based outlier rejection; otherwise the dip leaks into its own background and the map records roughly half the true depth.
- **Frame selection: bright first, then flat.** The dip is multiplicative, so signal scales with scene brightness. Selecting on flatness alone picks uniform overcast — the flattest sky there is, and exactly where the dip sits at the noise floor.
- **Edge-aware geometry.** The window is clamped to the frame — the blemish sits closer to the top edge than the window half-size — and the blemish’s local coordinates inside the window, plus the measured centroid of the map’s dip content, are published with the map. The corrector aligns the *content*, never the window origin.



(a) Measured radial profile vs. a Gaussian matched at half depth: the wings fall faster. Parametric fits locate; the empirical map corrects.



(b) Why offline reprocessing solves its gain in display space: with a fixed linear-space gain, the residual is proportional to the input dip.

Figure 3: Shape and gain measurements that fixed the design.

The publisher refuses to publish rather than publish junk: minimum frame counts per channel and a sanity range on the map’s peak depth. On any failure the live system keeps the previous good map.

## 4 Applying the correction

### 4.1 Position: three layers, fastest wins

The blemish migrates intraday, so placement cannot rely on the nightly fit alone. At every tick the corrector chains three estimators, each bounded by the one below it:

1. **Nightly posterior** (position, velocity, epoch) — the slow, validated anchor.
2. **Velocity extrapolation** from the fit epoch to now (fractional days), total displacement capped at 40 px.
3. **Persistent template lock** — the measurement update. The corrector cross-correlates the frame’s own dip (frame divided by its local background) against the map’s dip template within  $\pm 25$  px; a decisive lock (normalised correlation  $\geq 0.45$ , clean surrounding ring) snaps the map onto the measured position *and is saved*. A lock must also be unambiguous: the runner-up correlation peak (outside the winner’s neighbourhood) must trail by a margin, so a cloud edge or contrail that ties the blemish cannot capture the tracker. The next tick starts from the last saved lock (if fresh and within 40 px of the prediction) rather than from the prediction alone — a tick-cadence recursive filter whose prior is minutes old instead of a day old. The nightly prediction remains the fallback and the sanity bound, so one corrupted lock cannot walk the correction away. The published bundle itself is integrity-checked on load (schema version, array hash, finiteness); a torn or corrupt publish is rejected and the previous state persists.

The map content is placed by warping (bilinear, border transmittance 1.0) with the window origin fixed at its published, frame-clamped location — necessary because the blemish sits too close to the frame edge for the origin itself to follow it.

### 4.2 Amplitude: exposure scaling plus gated self-calibration

The map records depth at the reference exposure. Two mechanisms adapt it per frame:

- **Exposure scaling.** A measured depth-vs-level curve (fitted from within-tick bracket triples) scales the map’s deficit by  $s(L)/s_{\text{ref}}$ , where  $L$  is the frame’s local background level.
- **Per-tick self-calibration, gated.** The tone curve is uncalibrated, so on dark backgrounds the true dip runs deeper than the scaled map. When the surrounding ring is clean enough to trust (relative MAD  $< 0.05$ ), the corrector measures the frame’s own core deficit and trims the amplitude, clamped to  $[0.75, 1.35] \times$  the model. When the ring is cloudy the gate closes and the model amplitude stands — chasing a cloud-inflated “dip” is how inverse bright spots are made.

The correction itself divides rather than reconstructs:  $I \mapsto I/\tilde{T}$  with  $\tilde{T}$  feathered to unity outside the footprint. There is deliberately *no scene-referenced cap*: a clamp referenced to the observed image or a local sky fit eats the correction exactly where a bright cloud passes behind the blemish (the dimmed cloud becomes the reference), and an earlier pixelwise version also silently deleted the upper half of the noise distribution, leaving an unnaturally smooth disc. Safety is instead *model-referenced*: the correction’s shape is the quality-gated empirical map, and its amplitude is bounded relative to the characterized deficit — the closed-loop gain is clamped to  $[0.4, 1.8]$ , the live amplitude trim to  $[0.75, 1.35] \times$  the model with the exposure scale capped at 2.5, and the total applied deficit floored at  $T \geq 0.85$ . The correction can therefore never exceed a fixed multiple of what the blemish could physically account for, whatever the scene does; clouds neither evade the bound nor get eaten by it, and the noise texture passes through untouched at every frequency.

### 4.3 Offline reprocessing: closed-loop gain

For archive reprocessing, where four fusions per tick are affordable, the gain is solved in *display* space: fuse at adaptive probe gains bracketing the root, secant-step to the gain that nulls the *fused* dip, and refine once. This removes all dependence on the scene-dependent enhancement amplification (Figure 3b); the fitted gains (median 2.83) independently recover the measured  $\approx 2.85 \times$  CLAHE amplification.

## 5 Physics: what the data established

**The image’s motion is two-process — but the object need not move.** Step-level regression of daily blemish-image motion on scene drift (phase correlation on the building band) gives  $\Delta x_{\text{image}} \approx 0.93 \Delta x_{\text{scene}} + \epsilon$  with  $\text{med } |\epsilon| \approx 11 \text{ px/day}$ : camera rotation passes through at unit gain, plus an independent component — episodic, up to 27 px/day, continuous within days (7 px to 18 px, measured by splitting days into thirds). With the true shadow geometry (§9) the independent component requires no motion of the object at all: at  $\sim 3 \text{ cm}$  standoff the lever arm is  $\sim 92 \text{ px}$  of image shift per millimetre of camera–pane relative translation, so the entire week’s 120 px walk is 1.3 mm of cumulative flexure, and the intraday motion is 0.08 mm to 0.2 mm of thermal flex — unremarkable for a sun-struck sash with a camera resting against it. Day-by-day step ratios ranging from 1.06 to 26.7 simply reflect days when the mount flexed without rotating and vice versa. No constant ratio, and hence no feed-forward scheme, can track this; a near-field fiducial (a tape dot on the glass) would separate camera flexure from any residual object motion if the distinction ever matters. Operationally it does not: the corrector tracks the apparent position either way.

**A response-curve-confounded additive residual.** The slope of equation (1) is significantly nonzero in all channels and stable across nights (blue  $\approx +0.07$  at  $p \sim 10^{-9}$ ; green  $+0.022$ ; red  $+0.010$ ), strongly chromatic (B>G>R by  $\approx 6.7 \times$ , a Rayleigh-like ordering). Its sign implies a fixed *dark* offset — occlusion-like — rather than added scattered light. Three caveats mean this

is exploratory, not established: the photometry aperture was fixed at the original seed and drifted off the blemish for part of the record (it tracks the published centre as of 2026-07-28, so future nights measure the feature itself); the regression treats autocorrelated frames as independent; the linearisation is assumed inverse-sRGB, and a tone-curve error generates exactly this signature; and one early night shows the opposite sign. Until the response curve is calibrated per channel, the term should be read as a response-curve-confounded residual rather than physics. The correction is deliberately robust to the ambiguity: per-frame self-calibration corrects whatever depth each exposure actually exhibits, without needing to know why.

**Frame-edge geometry.** The blemish sits  $\sim 130$  px below the frame top while its background annulus extends to 170 px: the ring is partially clipped and the map window must be clamped (nothing can be cropped larger — the frame is the full sensor readout). Because the blemish is near-field, translating the camera up a few millimetres would move it  $\sim 100$  px down in frame while leaving the distant framing essentially untouched; that remains the physical fix for edge margin, with the caveat that migration continues regardless, so the tracker stays load-bearing.

## 6 System architecture: where each computation runs

The system is two-tier by design (Figure 4). Everything statistical and expensive runs *once per night* on a 32-core compute host (`akdeniz`); everything that runs *per tick* is closed-form, operates on the published  $300 \times 300$  window rather than the 4 K frame, and lives on the capture machine (`pandr`, an M1 MacBook Air) inside the corrector itself. Measured on real frames, the entire per-tick correction — placement, template lock, amplitude calibration, divide — costs 84 ms per bracket, about 0.25 s of a 15 s to 28 s capture cycle.

| level                                                | where                                     | cadence    | cost                                 |
|------------------------------------------------------|-------------------------------------------|------------|--------------------------------------|
| MAP position/shape fit (multi-start L-BFGS)          | <code>akdeniz</code>                      | nightly    | seconds                              |
| registered transmittance map ( $\sim 150$ frames)    | <code>akdeniz</code>                      | nightly    | $\sim 2$ min                         |
| acceptance validation (20 ticks $\times$ 4 fusions)  | <code>akdeniz</code>                      | nightly    | $\sim 4$ min                         |
| publish (map + model, $\sim 1.1$ MB)                 | <code>akdeniz</code> → <code>pandr</code> | nightly    | seconds                              |
| per-tick correction (place, lock, calibrate, divide) | <code>pandr</code>                        | every tick | 84 ms/bracket                        |
| archive reprocessing (closed-loop gain)              | <code>akdeniz</code>                      | on demand  | $\sim 7$ s/tick, $26\times$ parallel |

Table 1: Levels of computation. Only the nightly artifact crosses the network; the live path has no per-tick dependency on the compute host.

Two points about the split deserve emphasis. First, **the template lock is computed on the Air, not on `akdeniz`**: it is part of the corrector, executed per bracket on the local window — two small Gaussian blurs and one  $120 \times 120$ -in- $170 \times 170$  normalised cross-correlation, 5 ms to 10 ms. The compute host never sees a live frame; its entire influence on the live system is the nightly `spot_model.json` + `spot_tmap.npy` pair. After the MAP fit, nothing per-tick is an optimisation — a warp, a correlation, two 6-parameter least-squares fits, an elementwise divide.

Second, the split is about *availability* as much as cost: the corrector sits in the serving path, so a per-tick network dependency would turn any compute-host or network outage into a real-time degradation of the public site. As built, `akdeniz` can disappear for days: velocity extrapolation and the persistent lock carry the placement, staleness tiers step down gracefully, and the acceptance gate catches any quality regression when the nightly service resumes.

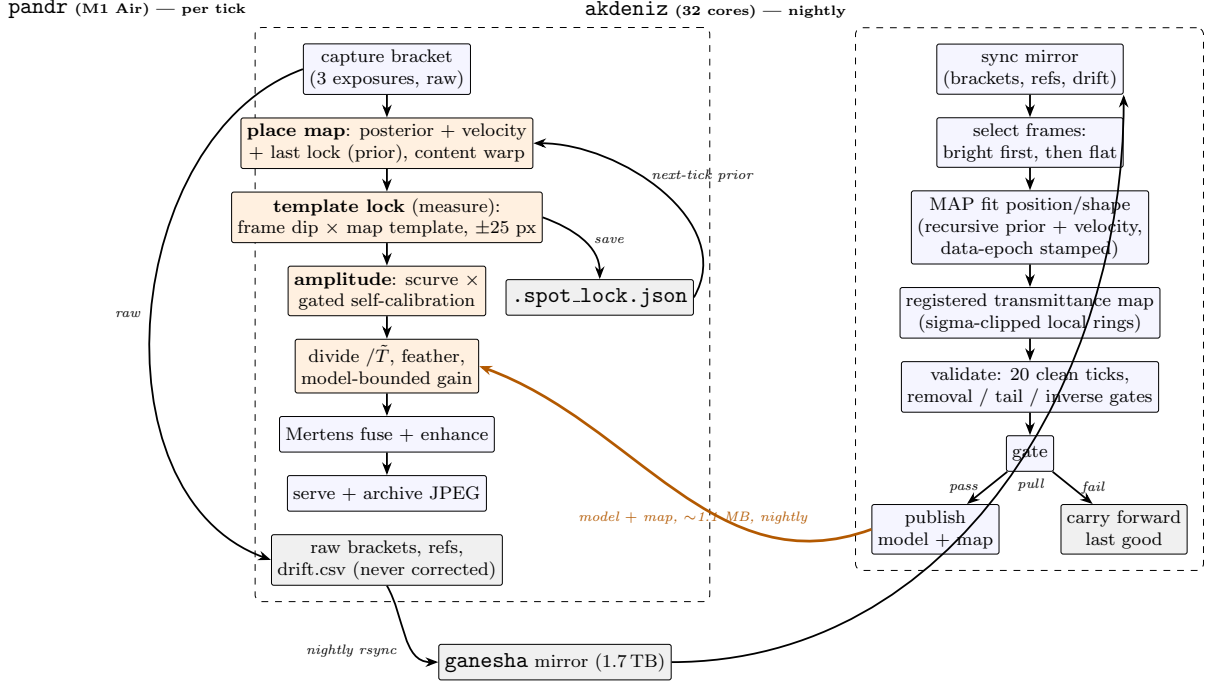


Figure 4: Data flow. Orange nodes run inside the corrector on the capture machine at every tick; the template lock and its persistence loop are local to the Air. The compute host’s only influence on the live system is the nightly published model+map (orange arrow); its only input is the raw-data mirror. Raw brackets and reference frames bypass the corrector entirely, so the measurement stack is never contaminated by its own correction.

## 7 Autonomous operation

A nightly job on **akdeniz** runs the whole cycle unattended: sync frames from the mirror; select frames; fit the position (recursive prior, pinned across brackets); build the registered maps; validate; publish to the capture host only if validation passes. Validation runs on clean-sky ticks — level and ring-structure thresholds measured *outside* the footprint, so the criterion cannot be circular — and the gate requires zero inverse spots, a floor on the median removal fraction, a bounded tail, and no regression against a rolling baseline of recent accepted nights. Failure paths are first-class: insufficient clear frames, low-confidence positions, and implausible jumps all carry the previous parameters forward with a log line, and repeated failures escalate. The corrector applies tiered staleness policies (beyond seven days it stands down to a gated fallback). Success is asserted from output artifacts, never process exit codes. Historical reprocessing uses the identical code path (`--replay`), which is also the honest test of autonomy: replaying the archive exercises cold start, carry-forward, gate rejections, and recovery exactly as a live outage would.

## 8 Performance

Live evidence is retained in machine-readable form (`spot_paper/evidence/live_paired.csv`: per tick, the served frame vs. a re-fusion of its own raw brackets uncorrected, with positions, levels, ring structure and model version). On the same clean-tick selection as Table 2: the clear evening of 07-27 gives  $n = 20$ , median removal 92.9 %, median served residual 0.35 %, range  $[-0.80, +1.44]$ ; 07-28 — a heavily clouded day during which the tracking chain was deployed and revised mid-day — gives  $n = 13$  post-deployment clean ticks with median removal 69.7 % and median residual 1.70 %. The honest reading is that clear-sky live performance matches the offline table and

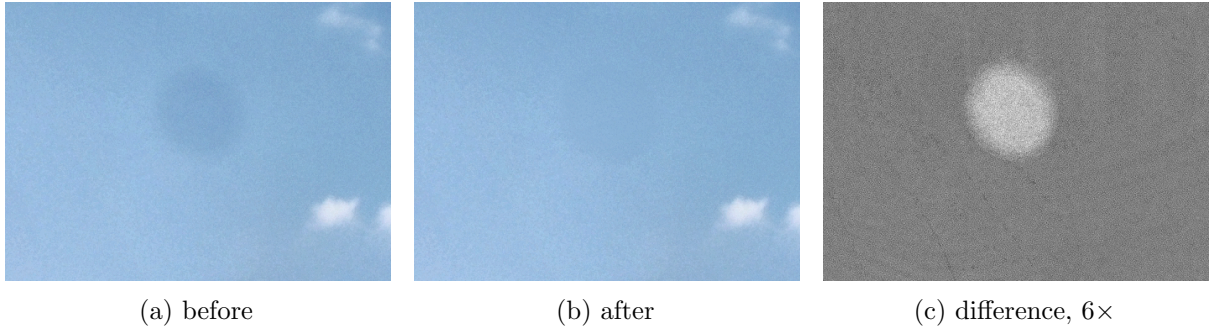


Figure 5: Pixel-scale detail of Figure 1 — the same three raw brackets through the same production chain (registered map, template lock, adaptive closed-loop gain, model-referenced amplitude bounds): 5.46 %  $\rightarrow$   $-0.5$  %. The difference (right, 6 $\times$ ) is a single compact blob; nothing outside the footprint is touched, and the noise texture inside it is preserved.

| stratum (offline, closed-loop gain) | $n$ | before | after  | removed | $ after  > 1.5\%$ |
|-------------------------------------|-----|--------|--------|---------|-------------------|
| clean daylight frames               | 495 | 4.46 % | 0.52 % | 88.1 %  | 8.3 %             |
| + ring-structure gate               | 301 | —      | 0.51 % | —       | 7.0 %             |

Table 2: Offline reprocessing statistics from `spot_fit_check/sweep2.csv`. Selection, disclosed: displayed frame level  $\geq 140$  and uncorrected measured dip in  $[1\%, 9\%]$  (row 2 adds ring-structure  $< 3$ ). The dip window *selects on a measured quantity*: it estimates performance on frames where a blemish-scale dip is detectable, and excludes negative, weak and cloud-inflated cases — it is not a generic daylight population. Residual distribution: 66 of 495 frames (13%) end negative, six below  $-0.5\%$ , one below  $-1\%$  (minimum  $-1.28\%$ ); “inverse bright spot” in the text means a residual below  $-1\%$ . The 495 frames span 8 days (11 to 111 per day); per-day median removal runs 83.2 % to 94.3 %, and a day-block bootstrap gives removal 88.1 % with 95% CI  $[85.6\%, 90.9\%]$  and median residual 0.52 % with CI  $[0.40, 0.66]$ .

cloudy-sky “clean” ticks remain contaminated by weather at the measurement aperture; a full clear-day out-of-sample evaluation is pending and the evidence file will accumulate it. Centring is monitored separately by a dipole statistic — the signed difference between the dip in the leading and trailing half-disks along the motion direction; a mis-centred correction is *stationary* and oriented, where cloud moves. After the epoch and measurement-update corrections this statistic fell from  $+2.5\%$  ( $n = 17$  frames) to  $+0.4\%$  ( $n = 2$ ), with the remainder symmetric weather signal.

## 9 What is the blemish? A physical identification

The answer required one datum that no amount of image analysis could supply: the camera’s actual aperture. Two successive wrong identifications — a migrating condensation droplet, then a reflection of a room object — were each internally consistent with an *assumed* webcam aperture of  $f/2$ – $f/2.4$ . The published specification for this camera is  $f/4.0$ , fixed focus at 30 cm to 120 cm: an entrance pupil of  $\approx 1.1$  mm, not 2.7 mm to 3.9 mm. With that single correction, every measurement converges on the mundane truth, which is also what direct inspection of the pane suggests: **a  $\sim 0.25$  mm speck of debris on the glass.**

**The solve.** A pixel viewing the distant scene collects a ray bundle that crosses the pane as a patch of diameter  $\approx a(1 - d/F)$ , where  $d$  is the pupil-to-pane distance and  $F \approx 0.5$  m the fixed focus distance. A speck of size  $s$  removes the fraction  $(s/a)^2$  of a pixel’s light and images with

full width  $\approx (s + a(1 - d/F))/d$ . The measured depth (3% to 7%) gives  $s \approx 0.22a \approx 0.24$  mm; the measured 76 px FWHM then requires  $d \approx 30$  mm to 35 mm. Of that, at most  $\sim 13$  mm is air gap and glass, implying the entrance pupil sits  $\sim 2$  cm behind the front of the lens — deep, but characteristic of short-focal-length inverted-telephoto designs, and a checkable prediction. The faster-than-Gaussian wings (Figure 3a) are the compact-support signature of exactly this: a hard-edged occcluder convolved with a hard-edged pupil. The neutral colour (integrated deficits equal across channels to 8%; the red channel deeper but narrower, i.e. chromatic defocus, not pigment) fits nondescript grey debris. The two or three fainter compact features elsewhere on the pane are siblings: other motes.

**Why the image moves.** The same geometry that blurs the speck amplifies motion: image shift equals camera–pane translation divided by  $d$ , about 92 px/mm. The feature’s entire week of “migration” (120 px) is 1.3 mm of accumulated flexure; its intraday drift is 0.08 mm to 0.2 mm — the thermal breathing of a sun-struck sash with a camera resting against it. Nothing on the glass ever moved. The kinematic anomalies that motivated more exotic identifications (episodic steps, opposite-signed step days, onset with the hot spell) are all natural behaviours of mm-scale structural flexure seen through a  $30\times$  lever.

**The colour of the blemish.** The registered per-channel transmittance maps permit a measurement most defects never get: the speck’s extinction spectrum, in three bands. The subtlety is separating the *object’s* colour from the *optics’* colour — longitudinal chromatic aberration gives each channel a slightly different defocus blur, and a tighter point-spread function makes the same grey object look deeper and narrower in that channel. The discriminator is the *integrated* deficit,  $\int(1 - T) dA$ , which is invariant to blur:

|                           | blue ( $\sim 460$ nm) | green ( $\sim 540$ nm) | red ( $\sim 610$ nm) |
|---------------------------|-----------------------|------------------------|----------------------|
| core depth                | 4.49 %                | 4.89 %                 | 5.38 %               |
| half-depth radius         | 51.4 px               | 50.7 px                | 48.8 px              |
| integrated deficit (rel.) | 1.00                  | 1.08                   | 1.07                 |

Red is 20% deeper than blue but also narrower, and the integrated deficits agree to within 8% across all three channels. The depth/width pattern is *not* lens chromatism — for this geometry every axial mechanism (normal dispersion, achromat secondary spectrum, edge diffraction at Fresnel number  $\sim 4$ , sensor carrier diffusion) predicts the red shadow *broad*er, the wrong sign. What produces deeper-and-narrower red at constant integral is the speck itself: micron-scale surface texture scatters short wavelengths at small angles, and scattered blue that stays within the pupil’s acceptance re-enters the bundles — refilling the shadow core (blue shallower), feathering its skirt (blue wider, by  $\sim 30\ \mu\text{m}$  at the pane), and slightly lowering the blue integral, exactly as measured. Red, barely scattered, casts the cleanest geometric shadow. The blur-invariant total extinction remains **neutral to  $\pm 8\%$** . That bound excludes strongly pigmented candidates: rust ( $\times$  several in the blue), organic tar or sap, pollen, chlorophyll debris. What remains is the unglamorous family that grey, wavelength-flat extinction describes: mineral dust, paint or plaster fleck, paper fibre, insect frass. In transmission at  $f/4$  the speck is, to the limit of what three colour channels can say, *grey* — consistent with the direct inspection finding an ordinary speck on the pane. (The blue-heavy *additive* residual of §5 is a separate, response-curve-confounded term and should not be read as the object’s colour.)

**Morals.** Three, earned expensively. First, *an assumed instrument parameter is a prior like any other, and a strong one*: two plausible physical stories were built downstream of  $f/2$ , and both collapsed the moment the real  $f/4$  was looked up. Second, the apparent motion of a near-field feature is a strain gauge, not a trajectory: at these standoffs the image measures *mechanics* (of

mount and sash) more than it measures the object. Third, the correction architecture was right to track the *apparent* position agnostically — every one of the candidate physical stories moved the image the same way, and the corrector never needed to know which was true. The remaining physical action item is pleasingly small: wipe the pane at the predicted spot (inner surface first, then outer — the 3 mm path difference changes the blur width measurably, so before-and-after even identifies which surface it was on), and keep the tracker running for the next mote.

## Code, data and reproducibility

Deployed code is byte-copied under `production/` with a component-status table in `PRODUCTION_STATUS.md`. Offline analysis code (Stan models, drivers, map builders, the intraday and colour measurements) is under `spot_paper/code/`; evidence files are `spot_fit_check/sweep2.csv` (offline table, filters as disclosed) and `spot_paper/evidence/live_paired.csv` (live paired measurements). The synthetic end-to-end test (a frame carrying exactly the published defect must null through the deployed corrector to  $< 0.5\%$ ) is `tests/test_synthetic_end_to_end.py`; the current chain passes at 0.000 %. A fuller test matrix (edge-clamped crops, injected motion, corrupted bundles, template-lock distractors) is specified in the 2026-07-28 review and remains future work.

## 10 Limitations

- Live post-deployment evidence is one mixed-conditions day; the offline table is the strong result, and live parity is demonstrated only for one clear evening.
- Cloudy-sky residuals at the blemish position are dominated by real cloud; the meaningful acceptance metric is clean-sky removal, and eye tests should distinguish stationary structure (defect) from structure that moves with the weather.
- The acceptance metric shares its centre with the estimator; a common-mode position error would pass unnoticed. The population median of the uncorrected dip is the cross-check — a mis-centred night reads visibly low.
- The nightly state model is position-plus-velocity; intraday migration is handled by the template lock, not modelled. If migration accelerates beyond the  $\pm 25$  px lock search radius within a day, tracking degrades gracefully to the prediction, and the gate catches the quality regression the next night.
- The frame-edge geometry worsens at  $\sim 2$  px/day; the camera lift is physical maintenance that software cannot substitute for.
- One-minute golden-hour frames keep no raw brackets, so their era corrections cannot be reprocessed; they are labelled rather than silently mixed with reprocessed frames.